Quantum Mechanics II — Exercise Sheet 1

Energy and position in the box

Variant: Solutions

Exercise 1 (Reconstructing an energy distribution). In the box $(0, L)$ a unit vector lies in $\operatorname{span}\{u_1, u_2, u_3\}$. Its mean energy is $5E_1$ and its second energy moment is $31E_1^2$. Find all three energy probabilities and the energy variance. State what these data do and do not determine about the vector.

Solution. Write $p_n = |\langle u_n, \psi\rangle|^2$. Normalization and the two moments give

$$p_1 + p_2 + p_3 = 1, \qquad p_1 + 4p_2 + 9p_3 = 5, \qquad p_1 + 16p_2 + 81p_3 = 31.$$

Subtracting the first equation from the second gives $3p_2 + 8p_3 = 4$. Subtracting the second from the third gives $12p_2 + 72p_3 = 26$. Solving gives

$$p_1 = \frac{1}{12}, \qquad p_2 = \frac{2}{3}, \qquad p_3 = \frac{1}{4}, \qquad (\Delta H)^2 = 6E_1^2.$$

The data determine the coefficient magnitudes but not their relative phases.

Exercise 2 (Position couples opposite parities). For $m, n \geq 1$ compute the matrix element $\langle u_m, Qu_n\rangle$. Treat $m = n$ separately. Use your result to decide which off-diagonal elements vanish. Here Q is multiplication by x in the box.

Solution. Using the normalized sine modes and setting $y = x/L$ gives

$$\langle u_m, Qu_n\rangle = 2L \int_0^1 y \sin(m\pi y) \sin(n\pi y)\, \mathrm{d}y.$$

For $m = n$, reflection about $1/2$ gives $L/2$. For $m \neq n$, the product-to-sum formula and integration by parts give

$$\langle u_m, Qu_n\rangle = \frac{L}{\pi^2}\left(\frac{(-1)^{m-n} - 1}{(m-n)^2} - \frac{(-1)^{m+n} - 1}{(m+n)^2}\right).$$

If m and n have the same parity both numerators vanish. If they have opposite parity the expression simplifies to

$$\langle u_m, Qu_n\rangle = -\frac{8Lmn}{\pi^2(m^2 - n^2)^2}.$$

Thus the off-diagonal elements vanish exactly for equal parities.

Exercise 3 (A sharp edge and finite energy). In the box let $\psi(x) = \sqrt{3/L}\, \mathbf{1}_{(0,L/3)}(x)$. (i) Find its sine coefficients and check their normalization by completeness. (ii) Decide whether the energy has a finite mean and whether $\psi \in D(H)$. Do not differentiate the discontinuous function as though it were smooth.

Solution. The coefficient is its inner product with the sine mode:

$$c_n = \frac{\sqrt{6}}{n\pi}\left(1 - \cos\frac{n\pi}{3}\right).$$

The position norm is $\|\psi\|^2 = (3/L)(L/3) = 1$. Completeness of the sine basis therefore gives $\sum |c_n|^2 = 1$. On the subsequence $n = 6j + 3$ the bracket equals 2, so

$$n^2|c_n|^2 = \frac{24}{\pi^2}.$$

The nonnegative series $\sum E_n|c_n|^2$ diverges. The energy mean is infinite, and the stronger domain condition $\sum n^4|c_n|^2 < \infty$ also fails.

Exercise 4 (An energy filter). A normalized box state has coefficients $c_n = C/n^3$ for $n \geq 1$, with $C > 0$. Find C, its mean energy and the probability that a reading is at least $9E_1$. Give the normalized vector after that reading is retained. You may use $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$ and $\zeta(6) = \pi^6/945$.

Solution. The normalization and the mean energy are

$$C = \zeta(6)^{-1/2}, \qquad \langle H\rangle = E_1\frac{\zeta(4)}{\zeta(6)} = \frac{21E_1}{2\pi^2}.$$

The retained outcomes have $n \geq 3$. Their total probability and conditional vector are

$$p = 1 - \frac{65}{64\zeta(6)}, \qquad \psi_{\text{kept}} = \frac{\sum_{n=3}^{\infty} n^{-3}u_n}{\sqrt{\zeta(6) - 65/64}}.$$

The original vector belongs to $D(H)$ because $\sum n^4|c_n|^2 = C^2\zeta(2) < \infty$.